

Activity 1.2.2 Converting Types of Motion

PLTW

CONVERTING TYPES OF MOTION



Mechanisms that Convert Motion



Exploring Motion Converting Mechanisms



Making Mechanisms



Conclusion

Mechanisms that Convert Motion

GOALS	MATERIALS	RESOURCES
• Apply your knowledge of simple machines to create complex mechanisms. • Understand how mechanisms are used to convert from one type of motion to another type of motion. • Create a physical prototype of a mechanism that converts one type of motion to another type of motion.		
GOALS	MATERIALS	RESOURCES
• VEX® V5 POE Kit		

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• <u>VEX® Parts</u> • <u>Activity 1.2.2 Converting Types of Motion (Downloadable PDF)</u>		

In the previous activity, you learned about the four types of motion. Many types of mechanisms convert motion from one type to another. Mechanisms are used in all types of products and processes, like automobiles and assembly lines.

Fran the Farmer uses mechanisms to process the eggs her chickens produce. Her hens' eggs travel through a series of conveyor belts as they are washed, sorted, spaced, stamped, and packaged.

The Egg Spacing Mechanism video shows the spacing stage of the egg sorting process on Fran's farm. The mechanism shown in the video lines up the eggs with uniform space between eggs.

Egg Spacing Mechanism video (The video has no sound.)

Reflection



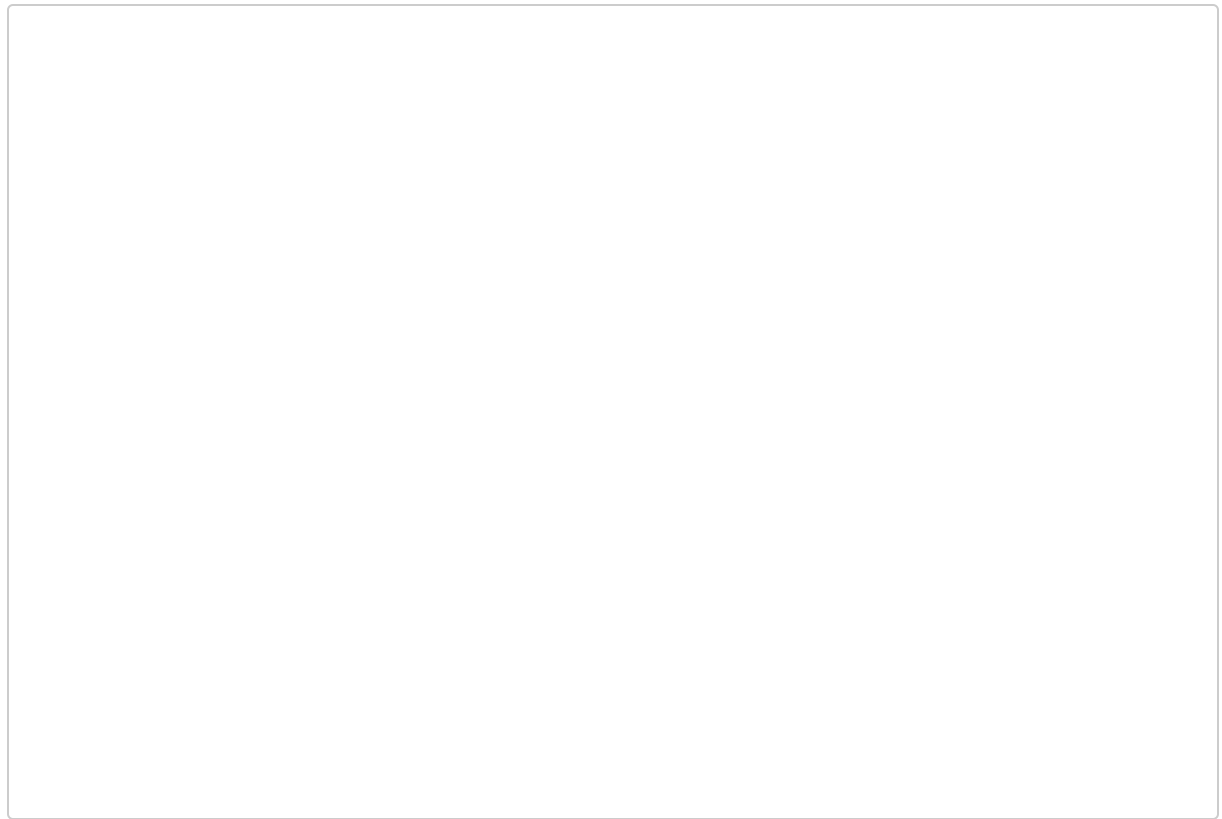
What kinds of mechanisms do you think are behind this part of Fran's egg-spacing process? What different types of motion do you see in the video?

Exploring Motion Converting Mechanisms

Interact with the five common mechanisms to view a video of each type of mechanism. Each of these mechanisms converts one type of motion to another type of motion. (The videos do not have sound.)

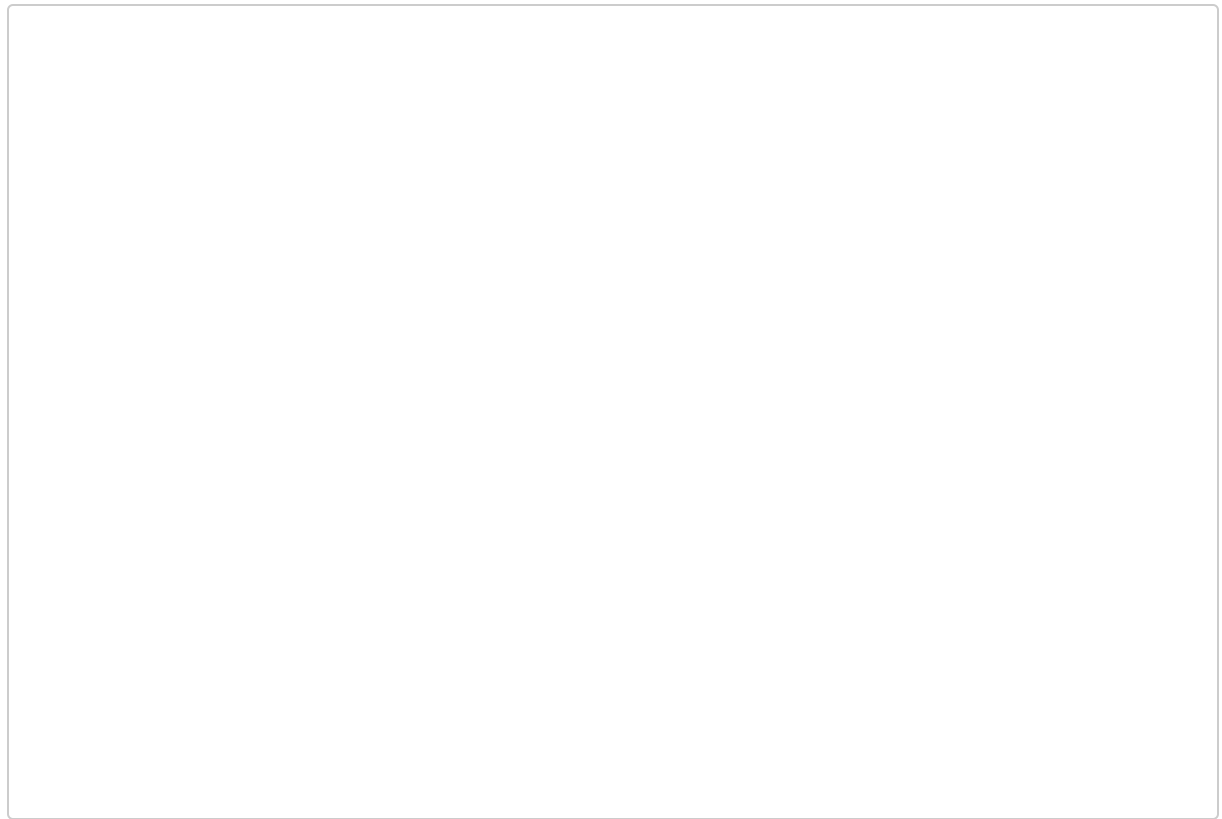
Rotary to Linear Motion (or Linear to Rotary Motion)

Rack and Pinion Mechanism video (The video has no sound.)



Rotary to Reciprocating Motion (or Reciprocating Motion to Rotary) —

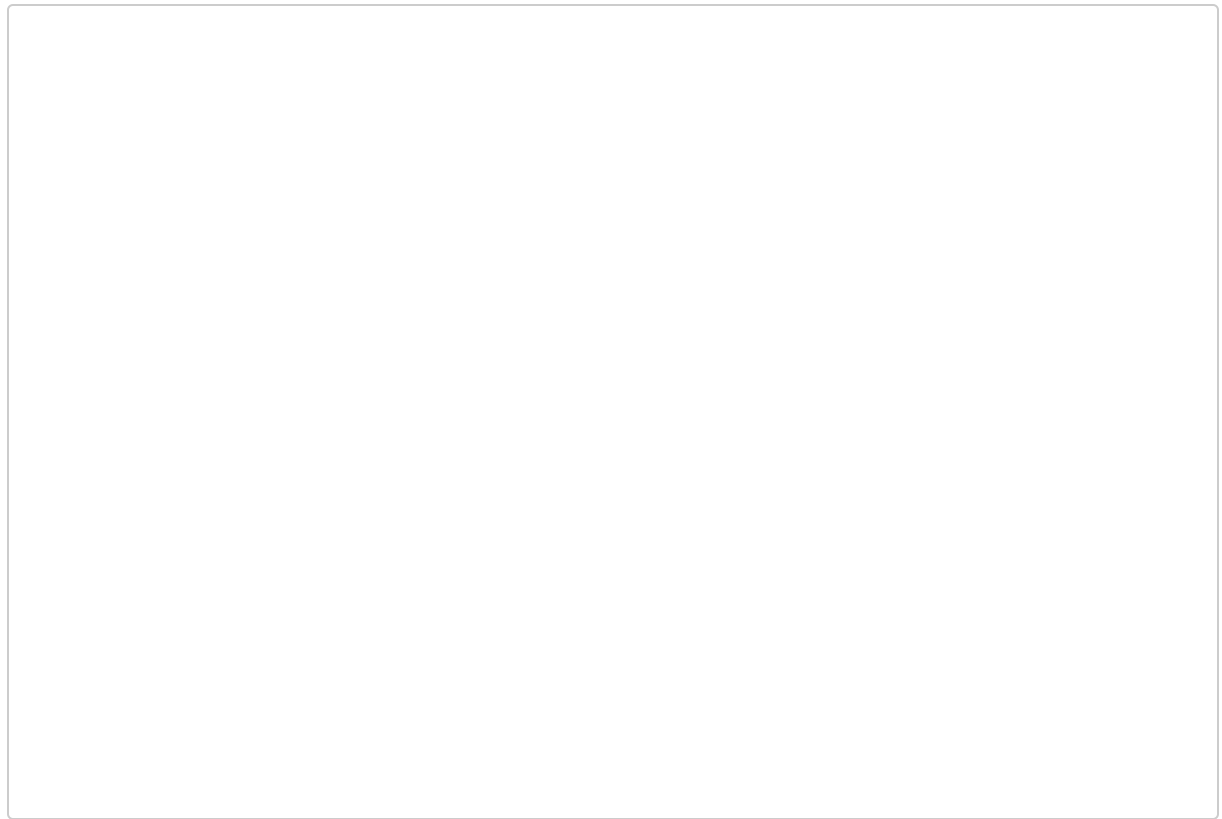
Crank and Slider Mechanism video (The video has no sound.)



Oscillatory to Rotary Motion (or Rotary to Oscillatory Motion) 1

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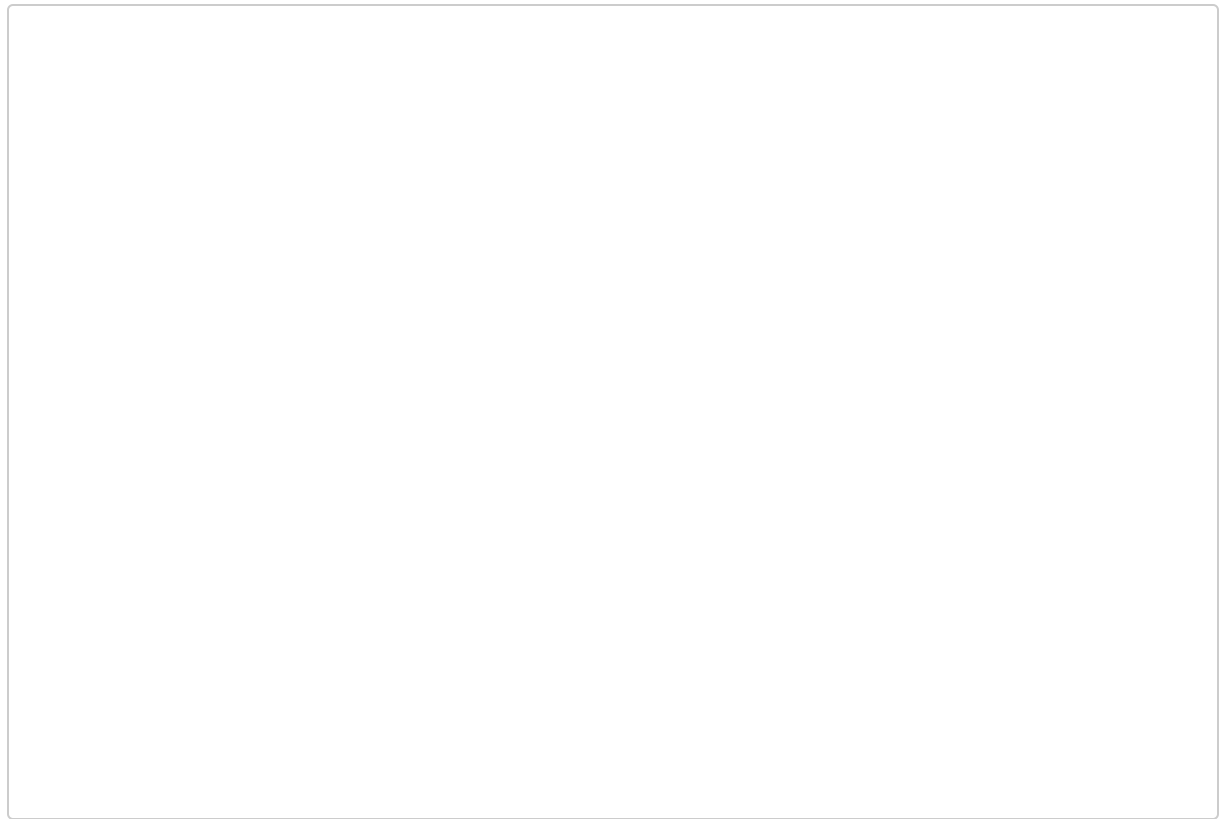
Treadle Linkage Mechanism video (The video has no sound.)



Oscillatory to Rotary Motion (or Rotary to Oscillatory Motion) 2

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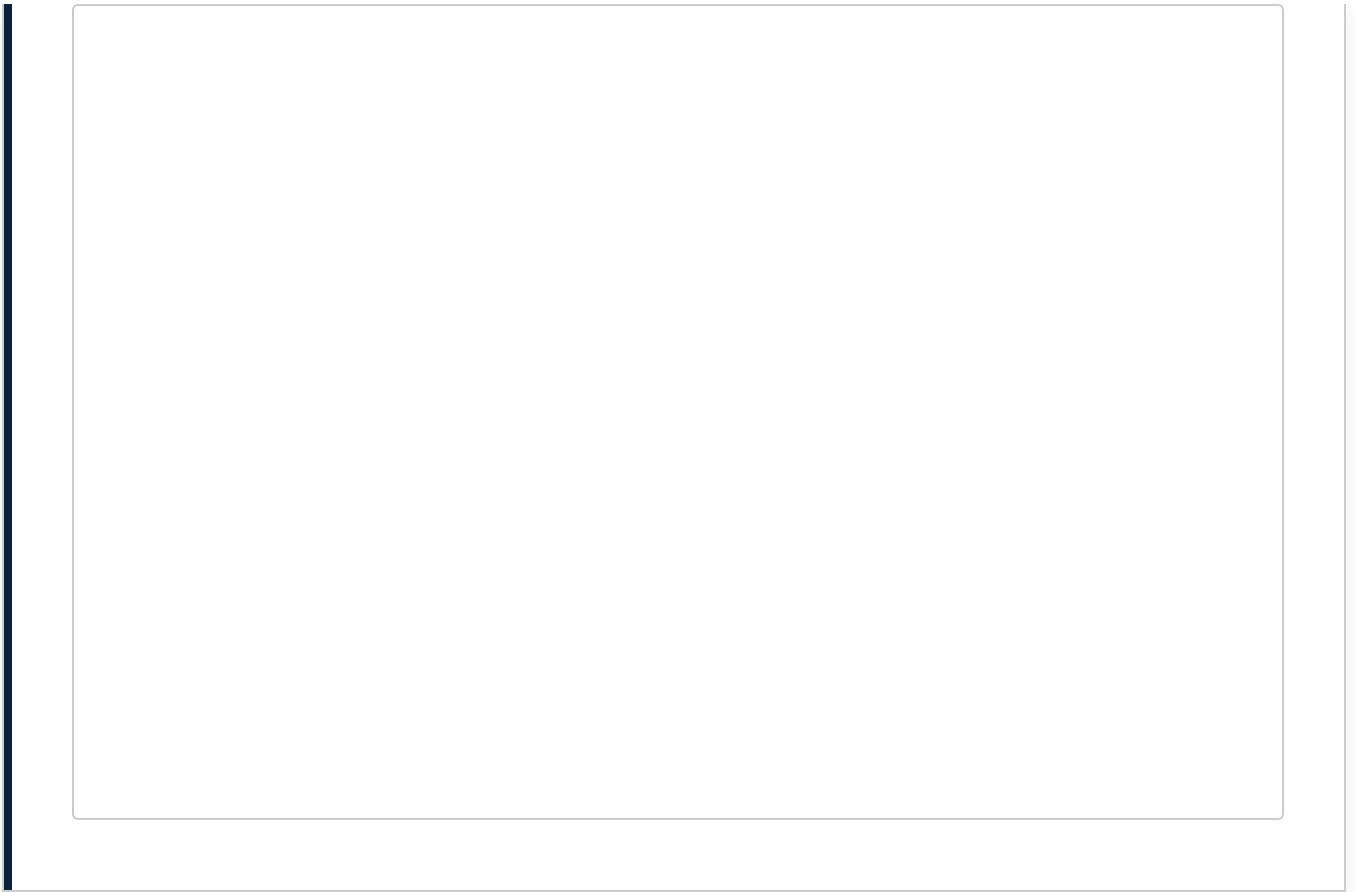
Quick Return Mechanism video (The video has no sound.)



Rotary to Reciprocating Motion —

One-Way Cam and Follower Mechanism video (The video has no sound.)

NOTE: The One-Way Cam and Follower can only be used to convert rotary to reciprocating motion, and not the other way (reciprocating to rotary motion) like many other mechanisms.



Turn and Talk



Where have you seen a mechanism like those shown above? If you haven't seen one before, where do you think these mechanisms could be useful?

Making Mechanisms

In this activity, you and a partner will use your VEX® V5 POE Kit to assemble a physical model of a mechanism that converts one type of motion to another.

1

Follow directions from your teacher for how to select a mechanism for this activity.

2

Research your mechanism. Find at least two real-world examples of how your mechanism is used in current engineering solutions.

3

Identify what types of motion your mechanism converts between.

HINT: Engineers typically consider four types of motion: reciprocating motion, linear motion, oscillating motion, and rotational motion.

4

Using **VEX® Parts**, assemble a physical model of your selected mechanism.

NOTE: You will not use motors for this activity. You will supply the input motion.

5

Share your VEX model with your classmates.

Conclusion

Question 1

What part of this activity did you find the most challenging? Why?

Question 2

What mechanism is present in the elliptical exercise machine shown in the Elliptical Exercise Machine in Use video? What types of motion does the mechanism convert between?

Elliptical Exercise Machine in Use video (The video has no sound.)